Analysis of Longitudinal Data — ARMD Trial

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1. Introduction and data

The ARMD trial is a randomised multicentre study in patients with age-related macular degeneration. Each patient received either interferon alfa-2a (active) or placebo. The response is visual acuity, the number of letters read correctly on a chart, measured at baseline and then at 4, 12, 24, and 52 weeks — four post-baseline measurements per patient over one year. The question is whether the active treatment slows the loss of vision.

I follow the ARMD case study of Gałecki and Burzykowski (2013) and redo it on my own random subsample of 150 patients. Broadly: look at the data first, fit the mean structure with `gls`, then the mixed models, and finally settle on one and check it.

1.1 Personal dataset

The subsample was built using the last two digits of my seed value, 43, as the random seed. I drew 150 indices between 1 and 240 and kept the corresponding subjects from `armd.wide`. The long-format version comes from the `armd` data frame in the same package, which already excludes the baseline measurement of the response and drops subjects without post-baseline data.

```
set.seed(43)
idx <- sample(1:240, 150)
sel <- levels(armd.wide$subject)[idx]
armd.wide.s <- droplevels(armd.wide[armd.wide$subject %in% sel, ])
armd.s <- droplevels(armd[armd$subject %in% sel, ])
```

The wide subsample has 150 patients, 73 on placebo and 77 on active treatment. Two of them have no measurement after baseline, so the long working set is left with 148 subjects and 553 observations. The split between arms stays balanced, which is expected because the original trial was randomised.

The data loss is monotone in most patients. The dropout pattern is summarised below (X indicates a missing visit).

Pattern	—-	—X	-XX	X—	XXXX	other
n	121	15	3	1	2	8

Table 1. Missing-data patterns in the personal subsample (wide format, 150 subjects).

Most patients (121) have all four visits complete. The rest drop out, and almost all dropouts happen at the end of follow-up. There is one patient missing only the baseline. I treat the missingness as ignorable and fit the models by maximum likelihood, which uses all available records under the MAR assumption. This is the same pragmatic choice made in the book.

2. Exploratory analysis

Before any model it is worth looking at the data. Table 2 gives the mean visual acuity and the standard deviation at each visit, separated by arm.

	4 wk	12 wk	24 wk	52 wk
Placebo mean	53.7	53.0	51.0	45.2
Placebo sd	16.0	17.5	17.8	16.8
Active mean	53.1	50.5	48.4	40.5
Active sd	16.1	17.9	17.8	19.1

Table 2. Mean and standard deviation of visual acuity by visit and treatment arm.

Acuity falls over time in both arms. The disease progresses, nothing strange there. What is worth noting is the sd column: it goes from about 16 at week 4 to 18–19 at week 52. The variance is not constant, it grows. The model will have to deal with that.

The active arm sits somewhat below placebo at each visit, and the difference is largest at week 52. Whether that is real or noise is what the modelling has to resolve. By eye it is not much.

[Figure 1. Mean visual-acuity profiles by treatment arm.]

[Figure 2. Individual visual-acuity trajectories for 24 randomly chosen patients (grey = placebo, blue = active).]

Figure 2 shows individual profiles for a random sample of patients. The trajectories are more or less linear, but each one starts in a different place and falls at its own rate. Some stay flat, others plummet. This is the spread that the random effects will later capture.

3. Mean structure and marginal models

I start with the book's marginal approach, fitting the mean structure by generalised least squares before introducing random effects. The mean model is the same one Gałecki and Burzykowski use for ARMD:

```
visual ~ -1 + visual0 + time.f + time.f:treat.f
```

With no general intercept, the baseline `visual0` enters as a covariate, `time.f` gives a mean per visit, and the interaction `time.f:treat.f` lets the treatment change by visit. I kept this parameterisation exactly as in the book.

3.1 Variance and correlation

The first `gls` fit assumes constant variance and independent errors (`fm6.1`). The exploratory plots already told me that was wrong. The variance grows with time, so I add a power-of-time variance function `varPower(form = ~time)`, giving `fm9.2`. And the repeated measurements on the same eye are correlated, so I add a within-subject correlation structure, first compound symmetry (`fm12.1`) and then a general unstructured one (`fm12.3`).

The likelihood-ratio tests all come out clearly significant. The power variance improves on the homoscedastic fit (LR = 60.2, $p < 0.0001$). Adding compound symmetry on top is a big jump (LR = 166.4, $p < 0.0001$). Moving from compound symmetry to a fully general correlation is still significant (LR = 46.5, $p < 0.0001$), which suggests the correlation is not exactly equal across all pairs of visits.

The correlations from `fm12.3` confirm this. Closer visits correlate more than distant ones: week 24 to 52 is about 0.76 versus only 0.28 from week 4 to 52. Compound symmetry forces them all to be equal, so it was always going to lose here.

pair	estimate	95% CI
4–12 wk	0.55	(0.43, 0.66)
4–24 wk	0.44	(0.28, 0.57)
4–52 wk	0.28	(0.07, 0.47)
12–24 wk	0.63	(0.52, 0.72)
12–52 wk	0.50	(0.35, 0.62)
24–52 wk	0.76	(0.68, 0.83)

Table 3. Within-subject correlations estimated with the general gls model (fm12.3).

4. Linear mixed-effects models

The `gls` models describe the correlation but treat it as a nuisance parameter. Mixed models instead attribute it to subject-specific random effects, which I find easier to interpret. Same mean structure, random effects added over the subject.

4.1 Random intercept

Start simple, one intercept per patient (`fm16.1`, `random = ~1 | subject`). The between-subject variance comes out at 70.4, the residual at 71.1. That puts the ICC right around 0.50, so half the variation is between patients and not within. Keeping the random effect is therefore worthwhile.

4.2 Adding the variance function and a random slope

The random intercept still assumes constant residual variance, which I already know is false. Adding `varPower(form = ~time)` to the residuals gives `fm16.2` and improves the fit substantially (LR = 41.1, $p < 0.0001$). Then I let the slope in time vary between patients too, `random = ~1 + time | subject`, which is `fm16.3`. Another clear improvement (LR = 54.1, $p < 0.0001$). Given how spread out the individual slopes looked in Figure 2, letting the slope vary was the obvious step.

Table 4 collects the whole sequence. AIC and BIC both fall as the structure becomes richer, and the random-intercept-plus-slope model with the power variance, `fm16.3`, comes out best among the mixed models. The general `gls` `fm12.3` has a slightly lower AIC but spends more parameters on the correlation and gives no subject-level interpretation, so I settle on `fm16.3` as the final model.

model	df	logLik	AIC	BIC
fm6.1 gls, homoscedastic	10	-2136.5	4293.0	4336.0
fm9.2 + varPower(time)	11	-2106.4	4234.8	4282.1
fm12.1 + compound symmetry	12	-2023.2	4070.4	4121.9
fm12.3 + general correlation	17	-1999.9	4033.9	4106.9
fm16.1 random intercept	11	-2065.6	4153.2	4200.5
fm16.2 rand. int. + varPower	12	-2045.0	4114.1	4165.7
fm16.3 rand. int. + slope + varPower	14	-2018.0	4064.0	4124.2

Table 4. Model comparison across the marginal (gls) and mixed (lme) sequence. Lower AIC/BIC is better.

5. Final model

The final model `fm16.3` has fixed effects for baseline acuity, the four visit means, and the four treatment-by-visit interactions, with random intercept and slope per patient and a power-of-time residual variance. The variance components are: intercept sd 6.25, slope sd 0.26 (correlation 0.19 between the two), and a residual scale of 4.78 with power parameter delta = 0.127. The positive delta confirms that the residual spread grows with time.

The fixed-effect estimates are in Table 5. Baseline acuity is a strong predictor: one extra letter of baseline gives about 0.92 more letters of acuity later ($p < 0.0001$), which is no surprise. The visit terms show the decline. The interaction terms are all negative, meaning the active arm does slightly worse than placebo at each visit, but none reaches significance on its own except a week-12 term at the margin.

term	estimate		t	p

		std. error		
visual0	0.924	0.043	21.52	<0.0001
time 4 wk	2.906	2.573	1.13	0.260
time 12 wk	2.329	2.635	0.88	0.377
time 24 wk	-0.539	2.760	-0.20	0.845
time 52 wk	-7.012	3.187	-2.20	0.028
4 wk : active	-0.986	1.429	-0.69	0.490
12 wk : active	-3.105	1.656	-1.87	0.062
24 wk : active	-2.284	2.017	-1.13	0.258
52 wk : active	-4.871	3.056	-1.59	0.112

Table 5. Fixed-effect estimates of the final model fm16.3.

5.1 Is there a treatment effect?

Looking at the interaction terms one by one is not the right test, because the treatment effect is spread across four coefficients. So I refit the model by maximum likelihood and compare it against a reduced model with no treatment terms at all, using a likelihood-ratio test. The result is LR = 4.33 on 4 degrees of freedom, p = 0.36. The treatment terms can be dropped. There is no evidence that interferon alfa-2a changes the course of visual acuity in this subsample.

This fits what is known about the trial. Interferon turned out not to help ARMD patients, and the book reaches the same conclusion. So the negative result here is reassuring rather than disappointing.

The marginal correlation implied by `fm16.3` still decreases with temporal distance, and the marginal variance grows from about 75 at week 4 to 315 at week 52, which matches the gls estimates and the raw standard deviations of Table 2. The random structure reproduces the pattern in the data.

6. Model checking

A few residual plots. Figure 3 shows the Pearson residuals against fitted values, split by visit. Centred on zero, and the spread is now roughly even across visits, so the power variance does its job. A few large negative residuals — patients who fell harder than predicted — but with no pattern.

[Figure 3. Pearson residuals versus fitted values, by visit.]

[Figure 4. Normal Q–Q plot of the Pearson residuals (left) and of the predicted random effects (right).]

Figure 4 shows the Q–Q plots. The residuals are fine in the centre, somewhat heavy in the tails from those same sharply-declining patients. The random effects look fairly normal. For clinical data this is acceptable, and I did not chase the tails with anything more elaborate.

7. Conclusions

In my subsample of 150 ARMD patients, acuity falls over the year, the variance rises as it progresses, and the repeated measurements are fairly correlated within each patient. The gls part confirmed all of that even before reaching the random effects.

The final model is `fm16.3`: random intercept and slope per patient plus a power-of-time residual variance. Half the variation is between patients, which is the main reason the random effects earn their place. The treatment effect, on the other hand, is not significant (p = 0.36). Interferon alfa-2a does not slow the loss of vision here, the same as the original trial and the book found.

Limitations: the dropout, for which I rely on the MAR assumption, and a handful of patients who fall much faster than the model likes.

Appendix. R code

Full code used to generate the data and the analysis, including the block that builds the personal dataset.

```r
# pec3 - longitudinal analysis armd
library(nlmeU)
library(nlme)
library(lattice)

# personal dataset
set.seed(43)
idx <- sample(1:240, 150)
data(armd.wide)
data(armd)
sel <- levels(armd.wide$subject)[idx]
armd.wide.s <- droplevels(armd.wide[armd.wide$subject %in% sel, ])
armd.s <- droplevels(armd[armd$subject %in% sel, ]) # long format, post-baseline only

# description
dim(armd.wide.s)
table(armd.wide.s$treat.f)
table(armd.wide.s$miss.pat)
table(armd.s$time.f)

# means and sd by time and treatment
with(armd.s, round(tapply(visual, list(time.f, treat.f), mean, na.rm=TRUE), 2))
with(armd.s, round(tapply(visual, list(time.f, treat.f), sd, na.rm=TRUE), 2))

# mean structure (gls)
lm1.form <- formula(visual ~ -1 + visual0 + time.f + time.f:treat.f)

# homoscedastic
fm6.1 <- gls(lm1.form, data = armd.s)

# power-of-time variance
fm9.2 <- gls(lm1.form, weights = varPower(form = ~time), data = armd.s)
anova(fm6.1, fm9.2)

# add within-subject correlation
fm12.1 <- gls(lm1.form, weights = varPower(form = ~time),
              correlation = corCompSymm(form = ~tp | subject), data = armd.s)
fm12.3 <- gls(lm1.form, weights = varPower(form = ~time),
              correlation = corSymm(form = ~tp | subject), data = armd.s)
anova(fm9.2, fm12.1, fm12.3)
intervals(fm12.3, which = "var-cov")$corStruct

# mixed models (lme)
# random intercept
fm16.1 <- lme(lm1.form, random = ~1 | subject, data = armd.s)
VarCorr(fm16.1)

# random intercept + power variance
fm16.2 <- lme(lm1.form, random = ~1 | subject,
              weights = varPower(form = ~time), data = armd.s)
anova(fm16.1, fm16.2)

# random intercept + slope + power variance
fm16.3 <- lme(lm1.form, random = ~1 + time | subject,
              weights = varPower(form = ~time), data = armd.s,
              control = lmeControl(opt = "optim", maxIter = 200, msMaxIter = 200))
anova(fm16.2, fm16.3)
summary(fm16.3)
VarCorr(fm16.3)

# implied marginal correlation
getVarCov(fm16.3, individual = "2", type = "marginal")

# treatment effect test (ML)
fm.full <- update(fm16.3, method = "ML")
```

```r
fm.red <- update(fm.full, fixed = visual ~ -1 + visual0 + time.f)
anova(fm.red, fm.full)

# diagnostics
plot(fm16.3, resid(., type = "pearson") ~ fitted(.) | time.f, abline = 0)
qqnorm(fm16.3, ~resid(., type = "pearson"))
qqnorm(fm16.3, ~ranef(.))
```